Hardware Fault Report

Workstation CPU instability — Intel Core i9-14900K

Date: 2026-06-27 | For: IT / workstation support | Method: automated Windows Event Log analysis

1. Summary

This workstation crashes under sustained heavy CPU load: full system crashes (BSOD) and frequent native process faults (illegal-instruction / segmentation fault). Windows Event Log evidence points to the processor — an Intel Core i9-14900K — exhibiting the well-documented Intel 13th/14th-generation “Raptor Lake” instability defect.

The Intel microcode mitigation (revision 0x12B) is ALREADY installed, yet illegal-instruction crashes persist under load — which indicates the CPU has likely already degraded and is a warranty replacement (RMA) candidate.

2. System configuration

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CPU         Intel Core i9-14900K  (24 cores / 32 threads)
Motherboard ASUSTeK PRIME Z790-P WIFI
BIOS        American Megatrends, version 1666, released 2024-09-27
Microcode   0x12B  (Intel Raptor Lake instability mitigation — PRESENT)
Memory      32 GB DDR5 (Kingston KF552C40-32), running JEDEC 4800 MT/s, XMP/overclock OFF
OS          Windows 11 (build 26100)
```

3. Symptoms observed

- Two full system crashes (BSOD) on 2026-06-27 during heavy multi-core compute, each rebooting the machine unexpectedly (Kernel-Power Event 41 / 6008); crash dump written to C:\Windows\MEMORY.DMP:

```
16:03   Bugcheck 0x0000003B (SYSTEM_SERVICE_EXCEPTION)  arg 0xC000001D = ILLEGAL_INSTRUCTION
16:10   Bugcheck 0x0000001E (KMODE_EXCEPTION_NOT_HANDLED) arg 0xC0000005 = access violation
```

- Numerous user-space native faults during a long compute job: ~19 worker processes died with SIGILL (illegal instruction, exit 132) or segmentation fault (exit 139) over a single 14-hour run.
- Strongly load-dependent: stable at ~16 parallel worker processes, unstable at ~28. Short/light tasks complete fine; only sustained all-core load triggers the faults.

4. Diagnosis

The fault signature — illegal-instruction and access-violation crashes that appear only under sustained heavy load, on an i9-14900K — matches the Intel Raptor Lake (13th/14th-gen Core i9) instability issue. The illegal-instruction BSOD (0xC000001D) is the kernel-level form of the same fault seen in user processes: the CPU is computing incorrect results / faulting at high all-core voltage and current.

The Intel microcode fix (revision 0x12B) is already applied. That microcode prevents further voltage-related degradation but does NOT repair damage already incurred. Persistent illegal-instruction faults with 0x12B present therefore indicate the processor has most likely already degraded.

5. Ruled out

- Memory (RAM): running at JEDEC-standard 4800 MT/s with XMP/overclock disabled; 23 GB of 32 GB free during the workload — not a memory-overclock or out-of-memory problem. (A formal memtest is still advised to fully close this out.)
- Missing microcode patch: the 0x12B mitigation is already installed.
- Software: application crashes occurred as varied, non-deterministic native faults, and the OS-level BSODs are independent of any application.
- The recurring WHEA “corrected hardware error” log entries are PCI-Express link errors (AER on root port 0:6:0, ~1-2 per day) — a separate, minor issue unrelated to the CPU crashes.

6. Recommended actions (for IT)

1. Update the motherboard BIOS to the latest ASUS PRIME Z790-P WIFI release (newer than 1666 / Sep 2024), which carries Intel microcode beyond 0x12B.
2. In BIOS, load the “Intel Default Settings” / Intel baseline power profile (ASUS boards apply elevated voltages by default). Disable any MultiCore Enhancement / overclock.
3. Stress-test under sustained all-core load (~1 hour) while monitoring Event Viewer for new 0xC000001D / WHEA / bugcheck events and CPU temperatures.
4. If illegal-instruction faults persist after steps 1-2: initiate an Intel RMA for the CPU. Intel has extended the 13th/14th-gen warranty to 5 years for this issue; the bugcheck codes and crash history in this report are accepted as supporting evidence.
5. (Optional) Run Windows Memory Diagnostic (mdsched.exe) to formally clear RAM, and reseat / firmware-update the PCIe device at bus 0:6:0 to address the minor AER warnings.

Appendix — raw evidence (Windows Event Log)

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System / WHEA-Logger  Event 17 (corrected, PCIe AER): 2026-06-24..27, 1-2/day,
        component “PCI Express Root Port”, device bus 0:6:0 (VEN_8086 DEV_A74D)
System / Event 1001 BugCheck: 0x3B (0xC000001D...) @ 16:03; 0x1E (0xC0000005...) @ 16:10,
2026-06-27
System / Event 41 (Kernel-Power) + 6008 (unexpected shutdown): 2026-06-27 ~16:03 and ~16:10
Application / Event 1000: python.exe faulting in ucrtbase.dll (native abort), 2026-06-26
Kernel crash dump: C:\Windows\MEMORY.DMP
```